

MAE 184 Flight Simulation Techniques

Homework #3 Solution

Problem 3.1 (2 pts)

The table below provides data from the airplane manufacturer regarding engine power required for the Cessna 182 to maintain level flight at 6,000 ft altitude. Use the JSBSim trim script (c182_trim.xml) to find the level-flight equilibrium for each of the flight conditions listed in the table. Use the JSBSim trim results to create a plot with true airspeed [units: kts, nmi/hr] on the x-axis and engine brake horsepower [units: hp] on the y-axis. Show the manufacturer data as a line, and your JSBSim data as individual dots. Be sure to show a legend and axis labels on your graph. Submit your plot in BMP, JPG, or PNG format.

Manufacturer Data 182P Cruise Performance (weight = 2950 lb)

True Airspeed [kts]	Brake Horsepower [hp]	Fuel Flow [gal/hr]
117	113	9.4
122	122	10.0
127	131	10.6
131	140	11.3
135	147	11.9
138	156	12.6
141	166	13.3
144	175	14.0

MATLAB:

```
# manufacturer data
mdat = [
117 113 9.4
122 122 10.0
127 131 10.6
131 140 11.3
135 147 11.9
138 156 12.6
141 166 13.3
144 175 14.0];

m_ktas = mdat(:,1); # true airspeed [kts]
m_vfps = (6076/3600)*m_ktas; # true airspeed [ft/s]
m_bhp = mdat(:,2); # brake power [hp]
m_gph = mdat(:,3); # fuel flow [gal/hr]

# trim data from JSBSim
jdat = [
197 116 9.5
206 124 10.1
```

```

214 133 10.7
221 142 11.3
228 150 12.0
233 157 12.5
238 164 13.0
243 172 13.5];

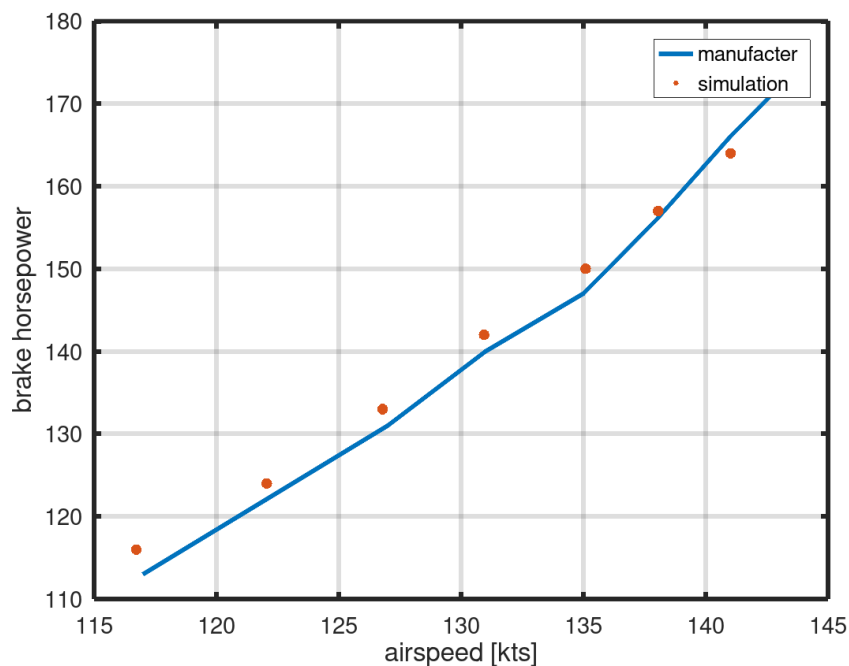
j_vfps = jdat(:,1); # true airspeed [ft/s]
j_ktas = (3600/6076)*j_vfps;
j_bhp = jdat(:,2); # brake power [hp]
j_gph = jdat(:,3); # fuel flow [gal/hr]

figure(1);
plot(m_ktas,m_bhp,"linewidth",3.0,j_ktas,j_bhp,".", "markersize",20.0)
set(gca,"linewidth",3.0,"fontsize",14);
ylabel("brake horsepower", "fontsize",16);
xlabel("airspeed [kts]","fontsize",16);
legend("manufacter","simulation")
grid on;

cp = 0.48; # guess bsfc [lb/hr/hp]
gph = cp*m_bhp/6.0;

figure(2);
plot(m_bhp,m_gph,"linewidth",3.0,m_bhp,gph,".", "markersize",20.0)
set(gca,"linewidth",3.0,"fontsize",14);
xlabel("brake horsepower", "fontsize",16);
ylabel("fuel flow [gal/hr]","fontsize",16);
legend("manufacter","cp fit")
grid on;

```



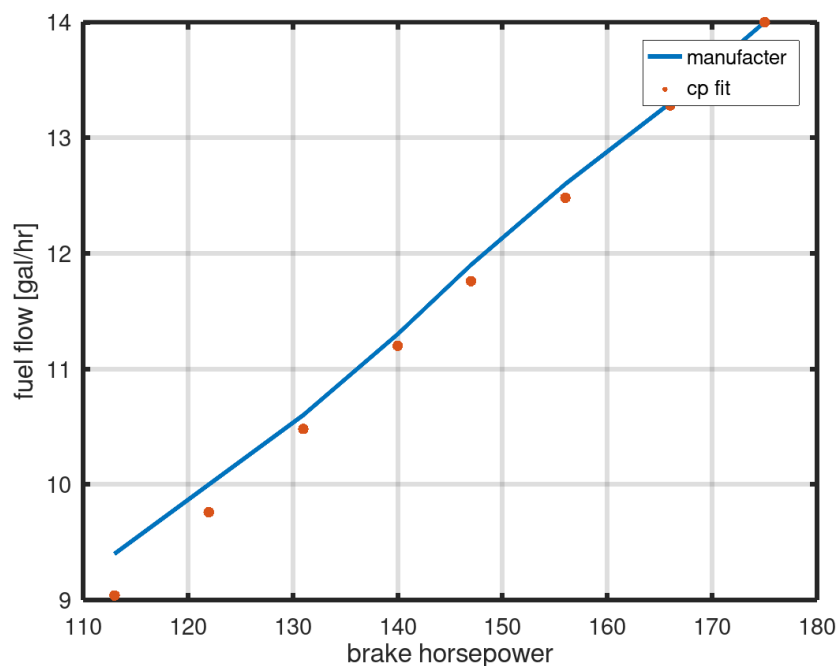
Problem 3.2 (2 pts)

Use the manufacturer fuel flow data from the previous problem to estimate the Brake Specific Fuel Consumption (BSFC) from the Cessna 182 model. Assume avgas fuel weighs 6.0 lbf/gal.

$$\frac{dW}{dt} = -c_p P \quad W = \text{weight}, P = \text{engine brake power}, t = \text{time}, c_p = \text{BSFC}$$

BSFC, c_p = _____ [lbf/(hr.hp)]

MATLAB: (see Problem 3.1 solution)



ANSWER: $c_p = 0.48$ [lbf/hr/hp]

Problem 3.3 (2 pts)

Use 1D linear interpolation to estimate the lift coefficient (CL) for an angle-of-attack of 11.0 degrees.

Angle-of-Attack (deg)	Lift Coefficient (CL)
8.0	0.7130
12.0	0.9450
14.0	1.0470
15.0	1.1330

- a. 0.771
- b. 0.829
- c. 0.887
- d. 0.945

MATLAB:

```
x = 11; # independent variable  
  
x0 = 8; # grid point below  
fx0 = 0.7130; # function at below grid  
  
x1 = 12; # grid point above  
fx1 = 0.9450; # function at above grid  
  
slope = (x - x0)/(x1 - x0);  
  
fx = fx0 + slope*(fx1 - fx0)
```

ANSWER: CL(11 deg alpha) = 0.887

Problem 3.4 (2 pts)

The fuel tanks of the Cessna 182 are located in the wings of the airplane. Estimate the body-axis moment of inertia I_{xx} contributed by the two fuel tanks. Assume the tanks are full of fuel and are shaped like a rectangular prism with dimensions given below.

fuel mass = 8.6 slugs (each tank)

fuel tank dimensions (each tank)

width (leading edge to trailing edge of wing) = 3 ft

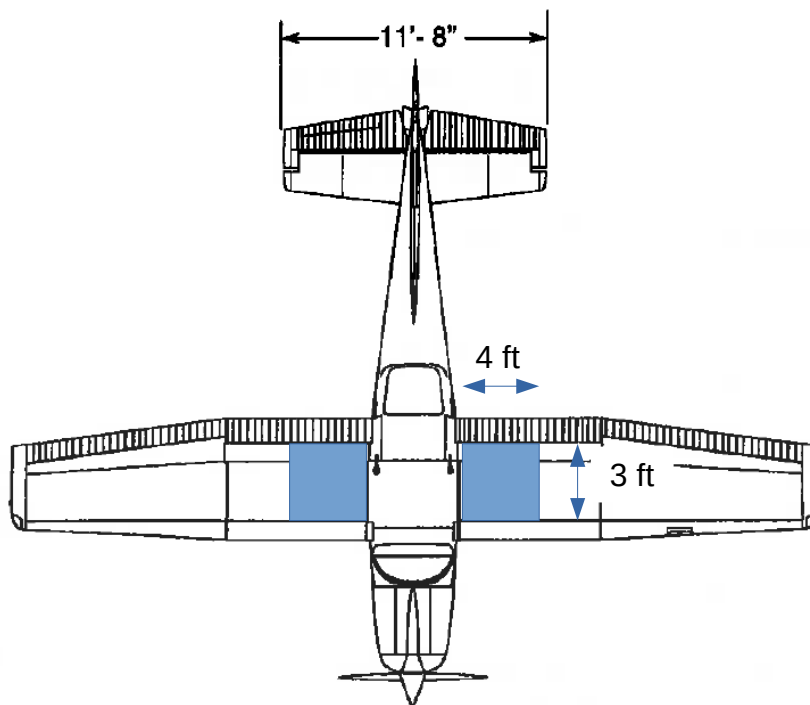
height (bottom of wing to top) = 0.5 ft

length (root of wing toward tip) = 4 ft

c.g. location in structural axis = (41.0, 0.0, 36.5) inches

right fuel tank center-of-mass in structural axis =
(48.0, 48.0, 59.4) inches

left fuel tank center-of-mass in structural axis =
(48.0, -48.0, 59.4) inches



I_{xx} (both tanks full) = _____ [slug.ft²]

MATLAB:

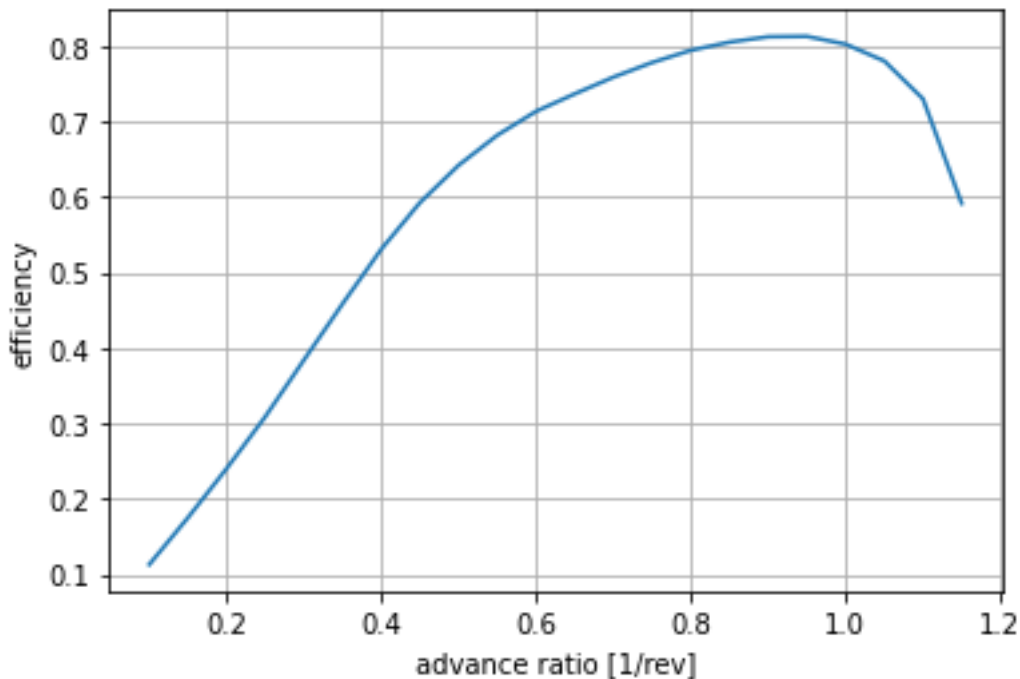
```
m = 8.6; % tank mass [slugs]
rcg = (1/12)*[41 0 36.5]'; % c.g in structural axis [ft]
TS2B = [-1.0 0.0 0.0; 0.0 1.0 0.0; 0.0 0.0 -1.0]; % rotate struc to body
Ixx = (m/12)*(4*4 + 0.5*0.5); % Ixx for tank about its c.m.
r1 = (1/12)*[48.0 48.0 59.4]'; % location of right tank [ft]
rb = TS2B*(r1 - rcg); % vector from c.g. to right tank [ft]
Ixx1 = Ixx + m*(rb(2)*rb(2) + rb(3)*rb(3)); % Ixx for right tank [slug.ft2]
r2 = (1/12)*[48.0 -48.0 59.4]'; % location of left tank [ft]
rb = TS2B*(r2 - rcg); % vector from c.g. to left tank [ft]
Ixx2 = Ixx + m*(rb(2)*rb(2) + rb(3)*rb(3)); % Ixx for left tank [slug.ft2]
Ixx = Ixx1 + Ixx2; % Ixx total both tanks [slug.ft2]

ANSWER: Ixx = 361 slug.ft^2
```

Problem 3.5 (2 pts)

Use the propeller efficiency curve shown below to estimate the thrust force [units: lbf] that the propeller is generating if the motor produces 130 hp. The 6.75 ft diameter propeller is rotating at a rate of 1800 rev/min and the aircraft is traveling at a true airspeed of 170 ft/s. (note: 1 hp = 550 ft.lb/s)

Thrust force = _____ [lbf]



MATLAB:

```
D = 6.75; # propeller diameter [ft]
bhp = 130; # engine brake power [hp]
V = 170; # airspeed [ft/s]
n = 1800; # propeller speed [rev/min]
n = n/60.0; # prop speed [rev/s]
```

```
J = V/n/D; # advance ratio [1/rev]
```

```
# look up prop efficiency
eta = 0.8; # prop efficiency
```

```
Thrust = eta*bhp*550/V # thrust [lb]
```

ANSWER: Thrust force = 336 [lbf]